Proposal

Bayesian F1 Grid Gap Analysis

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Packages

```
# install.packages("f1dataR")
# install.packages("dplyr")
```

Team

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Github Repo Link:

https://github.com/khuedo22/stat-405-final-project.git

Datasets

Primary Dataset

We use the `f1dataR` R package which pulls data directly from the Jolpica/Ergast F1 API. The dataset contains race results for every F1 season from 1950–2024. For our analysis we focus on the 2019–2024 seasons. Each row corresponds to one driver's result in one race.

URL: https://github.com/SCasanova/f1dataR

The key variables we use are:

- `driver_id` — unique identifier for the driver
- `constructor_id` — the team the driver races for
- `grid` — starting grid position (1–20)
- `gap` — finishing time behind the race winner in seconds
- `position` — finishing position
- `status` — whether the driver finished or retired (used for filtering)
- `circuit` — the circuit where the race took place (from schedule)
- `race_name` — the name of the Grand Prix (from schedule)

```
library(f1dataR)
library(dplyr)

# Load schedule to get circuit and race name
schedule <- load_schedule(season = 2023)

sample <- load_results(season = 2023, round = 1) |>
  mutate(
    season        = 2023,
```

```r
    round         = 1,
    race_name     = schedule$race_name[1],
    circuit       = schedule$circuit_id[1],
    top_speed_kph = as.numeric(top_speed_kph)
  )

head(sample)
```

```
## # A tibble: 6 x 16
##   driver_id constructor_id points position grid  laps  status gap   fastest_rank
##   <chr>     <chr>          <chr>  <chr>    <chr> <chr> <chr>  <chr> <chr>
## 1 max_vers~ red_bull       25     1        1     57    Finis~ 1:33~ 6
## 2 perez     red_bull       18     2        2     57    Finis~ +11.~ 7
## 3 alonso    aston_martin   15     3        5     57    Finis~ +38.~ 5
## 4 sainz     ferrari        12     4        4     57    Finis~ +48.~ 14
## 5 hamilton  mercedes       10     5        7     57    Finis~ +50.~ 10
## 6 stroll    aston_martin   8      6        8     57    Finis~ +54.~ 9
## # i 7 more variables: fastest <chr>, top_speed_kph <dbl>, time_sec <dbl>,
## #   season <dbl>, round <dbl>, race_name <chr>, circuit <chr>
```

Backup Dataset

Our backup is a Kaggle dataset containing F1 race results as pre-downloaded CSV files, available via our GitHub repository.

URL: https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020

The relevant files are `results.csv`, `races.csv`, `drivers.csv`, and `constructors.csv` which join on `raceId`, `driverId`, and `constructorId` respectively. Unlike the primary dataset, this source does not contain a `gap` column directly. Instead, we engineer it from the `milliseconds` column in `results.csv`, which records each driver's total race time in milliseconds. We compute the gap by subtracting the winner's race time from each driver's race time and converting to seconds:

```
gap = (driver_milliseconds - winner_milliseconds) / 1000
```

```r
base <- "https://raw.githubusercontent.com/khuedo22/stat-405-final-project/main/data_backup/"

results      <- read.csv(paste0(base, "results.csv"))
races        <- read.csv(paste0(base, "races.csv"))
drivers      <- read.csv(paste0(base, "drivers.csv"))
constructors <- read.csv(paste0(base, "constructors.csv"))

f1_backup <- results |>
  left_join(races |> rename(race_name = name),       by = "raceId") |>
  left_join(drivers,                                  by = "driverId") |>
  left_join(constructors |> rename(team_name = name), by = "constructorId") |>
  select(season = year, round, raceId, driverId,
         driver = surname, team = team_name,
         race_name, grid, position, points,
         milliseconds) |>
  filter(season == 2023) |>
  mutate(milliseconds = as.numeric(milliseconds)) |>  # force to numeric
  group_by(raceId) |>
  mutate(
    winner_ms = min(milliseconds, na.rm = TRUE),
    gap = (milliseconds - winner_ms) / 1000
```

```r
  ) |>
  ungroup()

head(f1_backup)
```

Question of interest

Question: Does grid position affect a driver's finishing gap to the race winner, and does this effect vary by circuit?

Grid position in F1 is set during qualifying: the faster you go, the further up the grid you start on race day. Starting further back generally means more cars to get past and a higher chance of getting stuck in traffic, which should translate to finishing further behind the winner. But how much this actually matters is probably not the same everywhere. At a place like Monaco, the streets are so narrow that overtaking is nearly impossible, so where you start is pretty much where you finish. Monza is the opposite: long straights and DRS zones mean cars can pass each other more freely, so a driver starting 10th still has a realistic shot at a top finish.

This leads to our question:

> *Does grid position affect a driver's finishing gap to the race winner, and does this effect vary by circuit?*

We want to estimate how much of a time penalty a driver faces for each grid position they start behind the front and whether that penalty is bigger at some circuits than others. We think Bayesian methods are a natural fit here since each circuit only has one race per season, so we are working with limited data per group. A hierarchical model lets circuits with less data borrow information from the broader pool of circuits rather than being estimated in isolation, which gives more reliable estimates especially for newer circuits on the calendar.

Methodology

First, we will preprocess the data and perform some EDA on the data, visualizing the relationship between grid positions and time gap for each circuit to understand the relationship.

The next step is to define some plausible models to use. A naive approach we may take is to assume all circuits have the same effect and model just the relationship between the grid position and the time gap. Since the time gap is strictly positive, we will model the log of the time gap + 1 to account for first place finishes (time gap of 0). The parameters for the distribution chosen below attempt to cover a wide range of plausible values since we don't know what parameter values are sensical.

$$
\begin{aligned}
\log(1 + \text{gap}_i) &\sim \mathcal{N}(\alpha + \beta \cdot \text{grid}_i,\ \sigma^2) \\
\alpha &\sim \mathcal{N}(0, 50) \\
\beta &\sim \mathcal{N}(0, 10) \\
\sigma &\sim \text{Exp}(1)
\end{aligned}
$$

Another option we could take is to model gap_i with an exponential distribution with parameter $\alpha + \beta * \text{grid}_i$.

β represents the average number of seconds behind the winner for each additional grid position. This would be the latent variable we are interested in.

To account for the difference in circuit, we can model this relationship for each 25 circuits. We can also model this in a hierarchical structure to reduce the sensitivity of the prior.

$$
\begin{aligned}
\log(1 + \text{gap}_i) &\sim \mathcal{N}(\alpha + \beta_j \cdot \text{grid}_i,\ \sigma^2) \\
\alpha &\sim \mathcal{N}(0, 50) \\
\beta_j &\sim \mathcal{N}(\mu_\beta,\ \sigma_\beta) \\
\sigma &\sim \text{Exp}(1) \\
\mu_\beta &\sim \mathcal{N}(0, 10) \\
\sigma_\beta &\sim \text{Exp}(1)
\end{aligned}
$$

j represents a circuit. We may compare the two models to understand the effect circuit has on the time gap.

We plan to compute the posterior of these models using MCMC in Stan and build credible intervals. After, we will perform diagnostics to evaluate the models, such as examining credible interval calibration via prediction, performing prior and posterior checks, and examining rank plots and trace plots. The results of these steps will determine our next steps, such as debugging slow mixing models or re-evaluating our model. This will also allow us to determine any limitations of our models. After building and checking our models, we will interpret our final results to see how grid position relates to the time gap and whether this effect varies by circuit. In addition to MCMC, we will refit the hierarchical model using Variational Inference and compare the two methods.

Tentative Plan

- literature review - Shahrukh
- problem formulation - Shahrukh and Khue
- data preprocessing - Shahrukh
- EDA - Khue
- Define two plausible models to test - Shahrukh and Khue (one each)
- run model in STAN - Shahrukh
- correctness check (software testing) - Shahrukh
- credible interval check - Khue
- diagnostics plot and other model evaluation - Khue
- model limitations - Khue
- interpretation - Shahrukh and Khue